FFT-based Orbit Response Matrix Analyzer

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Fermilab Accelerator Division
High-Level Application Suite

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Chapter 1

Introduction

1.1 What is FORMA?

FORMA (**F**FT-based **O**rbit **R**esponse **M**atrix **A**nalyzer) is a high-level application for measuring and analysing the *orbit response matrix* (ORM) of a circular particle accelerator. It drives corrector magnets with sinusoidal excitations, records beam-position monitor (BPM) readings, and extracts the linear transfer function between each corrector–BPM pair using spectral (FFT) analysis.

Orbit Response Matrix

The orbit response matrix $\mathbf{R}$ relates small changes in corrector strengths $\Delta\theta_j$ to the resulting changes in closed-orbit position Δx_i at each BPM:

$$\Delta x_i \;=\; \sum_j R_{ij}\,\Delta\theta_j \qquad \Longleftrightarrow \qquad \Delta\mathbf{x} = \mathbf{R}\,\Delta\boldsymbol{\theta}$$

Each element R_{ij} has units of *position per kick* (e.g. mm / A or mm / mrad).

1.2 Key Capabilities

- **AC excitation scans** — sinusoidal modulation of correctors, either simultaneously (fast) or sequentially (robust).
- **FFT-based extraction** — spectral isolation of the drive response from broadband noise.
- **Error estimation** — rigorous propagation of spectral noise into per-element uncertainties.
- **Safety monitoring** — real-time watchdog that aborts scans if orbit deviations exceed configurable thresholds.
- **Integrated analysis** — built-in heatmap display, plus an advanced viewer with SVD-based beta-function estimation, cosine-similarity analysis, and interactive slice plots with error bars.

1.3 Physics Background

In a circular accelerator the transverse closed orbit at location s_i satisfies

$$R_{ij} \;=\; \frac{\sqrt{\beta_i\,\beta_j}}{2\sin(\pi Q)}\,\cos\!\bigl(|\psi_i - \psi_j| - \pi Q\bigr), \tag{1.1}$$

where β_i, β_j are the Courant–Snyder betatron functions at the BPM and corrector, Q is the betatron tune, and ψ is the betatron phase advance. Measuring $\mathbf{R}$ therefore gives direct access to the optics of the machine.

1.3.1 Why AC Excitation?

A DC bump measurement (apply a static kick, measure the orbit change) is sensitive to any drift that occurs between the "bump-on" and "bump-off" readings. FORMA instead modulates each corrector at a known frequency and extracts only the coherent response at that frequency via the FFT. This rejects:

- slow drifts (appear at DC, which is excluded),
- 50/60 Hz mains ripple (appears at different frequencies),
- broadband BPM noise (spread across all bins, diluted by $\sqrt{N}$).

The result is a cleaner, higher-SNR measurement of R_{ij}, especially in a noisy operational environment.

Chapter 2

Mathematical Algorithms

This chapter gives a self-contained derivation of every formula used in FORMA.

2.1 Discrete Fourier Transform

Given a real-valued time series $x[n]$, $n = 0, \dots, N-1$, sampled at interval Δt, the DFT is

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}, \qquad k = 0, \dots, N-1. \tag{2.1}$$

The corresponding frequency of bin k is

$$f_k = \frac{k}{N\,\Delta t}, \qquad k = 0, \dots, \lfloor N/2 \rfloor - 1. \tag{2.2}$$

2.1.1 One-Sided Amplitude Spectrum

For a real signal, $X[k]$ and $X[N-k]$ are conjugate pairs. The one-sided amplitude spectrum (positive frequencies only) is defined as:

$$A[k] = \frac{2}{N}\, |X[k]|, \qquad k = 1, \dots, \lfloor N/2 \rfloor - 1. \tag{2.3}$$

The factor $2/N$ ensures that for a pure sinusoid $x[n] = A_0 \sin(2\pi f_0 n\,\Delta t)$, the peak amplitude $A[k_0]$ recovers the physical amplitude A_0, independent of N.

DC Component

The DC bin (k=0) is always excluded from analysis. Before computing the FFT the mean is subtracted from the signal: $s[n] = x[n] - \bar{x}$.

2.2 Drive Frequency Detection

For each corrector, FORMA finds the dominant excitation frequency by:

1. Computing the one-sided amplitude spectrum $A_c[k]$ (Eq. **??**).
2. Setting $A_c[0] = 0$ (ignore DC).
3. Selecting the bin with the largest amplitude:

$$k^* = \arg\max_{k \geq 1} A_c[k]. \tag{2.4}$$

4. Recording the drive frequency $f_c = k^*/(N\,\Delta t)$ and the complex Fourier coefficient

$$\tilde{A}_c = \frac{2}{N}\, X_c[k^*]. \tag{2.5}$$

2.3 Transfer Function and Response Matrix

For each (BPM i, corrector j) pair, FORMA computes the complex transfer function at the corrector's drive frequency:

$$H_{ij} = \frac{\tilde{A}_{b,i}[k_j^*]}{\tilde{A}_{c,j}[k_j^*]} = \frac{(2/N)\, X_{b,i}[k_j^*]}{(2/N)\, X_{c,j}[k_j^*]}, \tag{2.6}$$

where subscripts b and c denote BPM and corrector respectively. The $2/N$ factors cancel, but are retained for clarity. The response matrix element is the *real part* of this transfer function:

$$\boxed{R_{ij} = \mathrm{Re}(H_{ij}) = |H_{ij}| \cos \phi_{ij}} \tag{2.7}$$

where ϕ_{ij} is the phase of the transfer function.

Why take the real part?

For a quasi-static corrector kick (excitation frequency $\ll$ betatron frequency), the orbit response is in phase with the drive ($\phi \approx 0$ or π). Taking $\mathrm{Re}(H)$ preserves the *sign* of the response (positive = same direction as kick, negative = opposite), which is essential for orbit correction algorithms. The imaginary part, representing any phase lag, is discarded.

2.4 Noise Estimation

Accurate error bars require estimating the spectral noise floor of both the corrector and the BPM signals, excluding the deliberate drive signal.

2.4.1 Corrector Noise σ_c

After computing the corrector's amplitude spectrum, FORMA zeros out the drive peak and its two nearest neighbours on each side (a ± 2 bin exclusion zone):

$$\mathcal{E}_c = \{k^* - 2,\ \ldots,\ k^* + 2\}, \tag{2.8}$$

then computes the RMS of the remaining bins:

$$\sigma_c = \sqrt{\frac{1}{N_{\mathrm{bins}}} \sum_{k \notin \mathcal{E}_c,\, k \geq 1} |A_c[k]|^2}\,. \tag{2.9}$$

2.4.2 BPM Noise σ_b

BPM spectra contain peaks at *every* corrector's drive frequency (because all correctors affect all BPMs). The exclusion zone is therefore the union over all correctors. Additionally, frequency-to-bin mapping is done per-BPM using each BPM's own sample count and timing:

$$k_c^{(b)} = \mathrm{round}\big(f_c \cdot N_b \cdot \Delta t_b\big), \tag{2.10}$$

and the exclusion set becomes

$$\mathcal{E}_b = \{0\} \ \cup \bigcup_{\text{correctors } c} \{k_c^{(b)} - 2,\ \ldots,\ k_c^{(b)} + 2\}. \tag{2.11}$$

The BPM noise is then

$$\sigma_b = \sqrt{\frac{1}{N_b/2} \sum_{k \notin \mathcal{E}_b} |A_b[k]|^2}\,. \tag{2.12}$$

Why ± 2 bins?

Without windowing the FFT uses a rectangular window whose spectral leakage falls off only as $1/k$. The ± 2 exclusion zone catches the worst sidelobes of the drive peak. A wider zone or a Hann window would reduce leakage further at the cost of frequency resolution.

2.5 Error Propagation

The response matrix element is $R = \mathrm{Re}(Y/X)$ where $Y = \tilde{A}_b$ and $X = \tilde{A}_c$ are the complex Fourier coefficients of the BPM and corrector at the drive frequency.

Under the assumption of circular Gaussian noise in the complex plane, the per-component (real or imaginary) variance of a spectral coefficient is half the magnitude variance:

$$\mathrm{Var}[\mathrm{Re}(\tilde{A})] = \mathrm{Var}[\mathrm{Im}(\tilde{A})] = \frac{\sigma^2}{2}\,, \tag{2.13}$$

where σ is the magnitude RMS from Eq. **??** or **??**.

Applying standard error propagation to $R = \mathrm{Re}(Y/X)$, with $A_b = |Y|$ and $A_c = |X|$:

$$\boxed{\Delta R_{ij} = \sqrt{\frac{\sigma_b^2}{2\,A_c^2} + \frac{A_b^2}{2\,A_c^4}\,\sigma_c^2}} \tag{2.14}$$

where:

- σ_b = BPM noise RMS (Eq. **??**),
- σ_c = corrector noise RMS (Eq. **??**),
- $A_b = |\tilde{A}_{b,i}[k_j^*]|$ = BPM response amplitude,
- $A_c = |\tilde{A}_{c,j}[k_j^*]|$ = corrector drive amplitude.

Interpretation of the Two Terms

- **First term** $\sigma_b^2/(2A_c^2)$: uncertainty from BPM measurement noise. Larger corrector drives (A_c) suppress this term.
- **Second term** $A_b^2\sigma_c^2/(2A_c^4)$: uncertainty from corrector amplitude noise. Larger responses (A_b) amplify this term because a noisy denominator has a bigger effect.

2.6 Scan Waveform Generation

Each corrector is driven with a sinusoidal waveform superimposed on its nominal (operational) setpoint:

$$x_j[n] = x_{0,j} + A_j\,\sin\!\left(\frac{2\pi\,P_j}{S}\,(n \bmod S)\right), \qquad n = 0, 1, \ldots, N_{\text{total}}-1, \tag{2.15}$$

where:

- $x_{0,j}$ = nominal value (fetched from hardware),
- A_j = user-specified amplitude,
- P_j = number of periods for this corrector,
- S = points per superperiod,
- $N_{\text{total}} = S \times$ (number of superperiods) = total scan steps.

2.6.1 Simultaneous Mode

All correctors modulate in parallel during a single scan. Each corrector can have a different number of periods P_j, giving it a distinct frequency in the FFT and enabling separation of overlapping responses. This is N times faster than sequential mode.

2.6.2 Sequential Mode

One corrector modulates at a time; all others remain at their nominal values. After completing the full waveform, data is saved and the next corrector begins. This provides maximum SNR and eliminates cross-talk, but takes N times longer.

2.7 Beta-Function Estimation (SVD)

The analysis viewer estimates the betatron function at each BPM from the measured response matrix using singular value decomposition.

2.7.1 SVD Decomposition

The response matrix is decomposed as

$$\mathbf{R} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^T, \tag{2.16}$$

where $\mathbf{U} \in \mathbb{R}^{m\times k}$ (BPM structure), $\mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \dots, \sigma_k)$ (singular values), and $\mathbf{V}^T \in \mathbb{R}^{k\times n}$ (corrector structure), with $k = \min(m, n)$.

2.7.2 Beta Extraction

From Eq. **??**, each row of $\mathbf{R}$ is proportional to $\sqrt{\beta_i}$ (modulated by the cosine phase term). The dominant SVD mode captures this structure:

$$\hat{\beta}_i = \left(U_{i,1} \cdot \sigma_1\right)^2. \tag{2.17}$$

This estimate is then normalised so that the maximum value matches the maximum absolute element of $\mathbf{R}$:

$$\beta_i^{(\mathrm{norm})} = \frac{\hat{\beta}_i}{\max_i \hat{\beta}_i} \times \max_{i,j} |R_{ij}|. \tag{2.18}$$

Fallback Method

If the SVD fails (e.g. due to a degenerate matrix), the fallback estimate is the row-wise L2 norm: $\beta_i^{(\mathrm{fallback})} = \sqrt{\sum_j R_{ij}^2}$.

2.8 Cosine Similarity

The similarity matrix quantifies how alike two correctors' response patterns are. Let $\mathbf{c}_k$ be the k-th column (corrector) of $\mathbf{R}$, viewed as a vector in BPM space. The cosine similarity is:

$$S_{kl} = \frac{\mathbf{c}_k \cdot \mathbf{c}_l}{\|\mathbf{c}_k\|\,\|\mathbf{c}_l\|} \in [-1,\, 1]. \tag{2.19}$$

- $S_{kl} = +1$: correctors k and l affect BPMs identically (redundant).
- $S_{kl} = 0$: orthogonal response patterns (independent control).
- $S_{kl} = -1$: exactly opposite effects.

2.9 Safety Monitoring Algorithm

During a scan, the safety system monitors each BPM's position in real time using a rolling-average mean-shift test.

2.9.1 Baseline Establishment

Before scanning, N_0 baseline samples are collected (no excitation). For each BPM:

$$\mu_i^{(0)} = \frac{1}{N_0} \sum_{n=1}^{N_0} x_i^{(0)}[n], \qquad \sigma_i^{(0)} = \sqrt{\frac{1}{N_0} \sum_{n=1}^{N_0} \left(x_i^{(0)}[n] - \mu_i^{(0)}\right)^2}. \tag{2.20}$$

2.9.2 Real-Time Checking

A circular buffer of size B stores the most recent readings. The deviation metric is the *mean shift in units of the baseline standard deviation*:

$$n_{\sigma,i} = \frac{\left|\bar{x}_i^{(\text{current})} - \mu_i^{(0)}\right|}{\sigma_i^{(0)}}, \qquad \bar{x}_i^{(\text{current})} = \frac{1}{B} \sum_{k=1}^{B} x_i[k]. \tag{2.21}$$

2.9.3 Threshold Logic

$$\text{Action} = \begin{cases} \textbf{ABORT} & \text{if } n_\sigma \geq n_\sigma^{(\text{abort})}, \\ \textbf{WARNING} & \text{if } n_\sigma^{(\text{warn})} \leq n_\sigma < n_\sigma^{(\text{abort})}, \\ \textbf{SAFE} & \text{if } n_\sigma < n_\sigma^{(\text{warn})}. \end{cases} \tag{2.22}$$

An **ABORT** immediately halts the scan and restores all correctors to their nominal values.

Chapter 3

Getting Started

3.1 System Requirements

Component	Requirement
Python	3.9 or later
Operating system	Windows (primary), Linux (supported)
Display	1280×800 minimum
Network	Access to ACNET control system
Authentication	Valid Kerberos principal

3.2 Installation

1. Ensure all dependencies are installed:

   ```
   pip install -r requirements.txt
   ```

   The key packages are: `numpy`, `pandas`, `matplotlib`, `scipy`, `PyQt5`.
2. The `acsys` module must be installed separately from Fermilab sources.
3. `tkinter` is included with most Python distributions.

3.3 Launching the Application

3.3.1 Windows (Batch File)

Double-click `RUN_FORMA.bat`, or from a command prompt:

```
cd \path\to\FORMA
RUN_FORMA.bat
```

3.3.2 Direct Python

```
python forma.py
```

3.3.3 What Happens at Startup

1. The main window opens with nine tabs (all initially empty).
2. Authentication status shows **Not Authenticated** (red).
3. Safety banner shows **NOT CONFIGURED** (yellow).
4. The Activity Log tab begins capturing all console output.

Chapter 4

Complete Workflow

This chapter walks through the typical end-to-end usage of FORMA, from authentication to final analysis. A visual overview is shown in Figure ??.

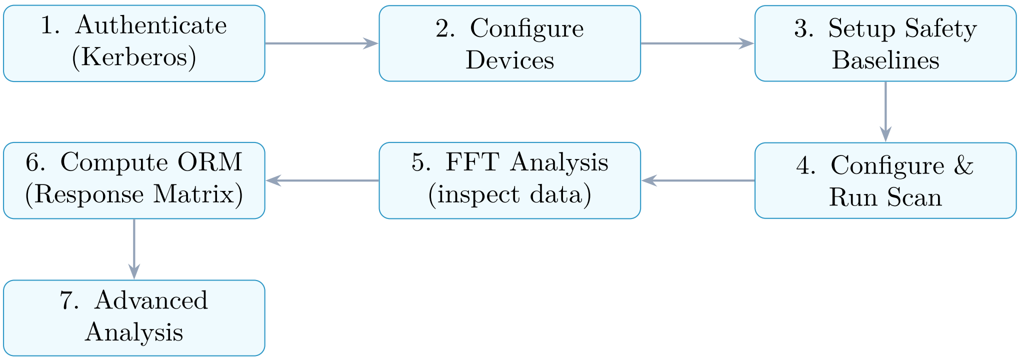

Figure 4.1: Typical FORMA workflow.

Chapter 5

Tab-by-Tab Reference

5.1 Tab 1: Device Settings & Selection

This tab is the starting point for every session.

5.1.1 Authentication

Step: Log in to Kerberos

Click `Kerberos Login` (or use `Authentication > Kerberos Login`). Enter your Kerberos principal and password in the dialog. On success the status changes to **Authenticated** (green).

Authentication Required

Most hardware-interacting features (fetching nominals, applying settings, running scans) require valid Kerberos credentials. Always authenticate first.

5.1.2 Adding Devices

There are two methods:

Manual Entry

1. Type a device name in the *Device Name* field (e.g. `H:CORR01`).
2. Click `Add to Setting List` (for correctors) or `Add to Reading List` (for BPMs).
3. Repeat for each device.

File-Based Selection

1. Click `Open Device Selection Tool...`
2. Select an Excel (`.xlsx`/`.xls`) or CSV file containing device names in the first column.
3. A dialog shows the available sheets; select the ones to import.
4. Devices are added to the appropriate list.

The status labels at the bottom show the current count: e.g. "*12 devices in Setting List.*"

5.1.3 Device Settings Grid

After adding setting devices, a scrollable grid appears showing each device with an editable *New Value* field. This lets you set custom setpoints before a scan if needed.

5.1.4 Control Buttons

Button	Action
`Fetch Nominals`	Reads the current operational values of all setting devices from ACNET.
`Apply Settings`	Writes the values in the *New Value* fields to hardware (with confirmation dialog).
ACNET Role	Dropdown to select the control role: `OPERATOR`, `testing`, `linac_trims`, `linac_quads`.
`Clear Setting List` / `Clear Reading List`	Removes all devices from the respective list.

5.2 Tab 2: Scan Settings

This tab configures scan parameters and launches the measurement.

5.2.1 Global Parameters

Parameter	Default	Description
Points per Superperiod	100	Number of sample steps in one modulation cycle. More points give finer frequency resolution in the FFT.
Number of Superperiods	1	How many complete cycles to execute. More cycles reduce noise by $\sqrt{N_{\mathrm{sp}}}$.

5.2.2 Scan Mode

Mode	Description
Simultaneous (default)	All correctors modulate in parallel. Fastest option ($N\times$ speed-up). Each corrector should use a *different* number of periods to separate their frequencies.
Sequential	One corrector at a time. Maximum isolation and SNR but $N\times$ slower.

5.2.3 Per-Device Settings

The treeview table shows one row per setting device with editable columns:

Column	Meaning
Device	Device name (read-only).
Amplitude	Peak amplitude of sinusoidal excitation (in device units).
Number of Periods	How many full sine cycles within one superperiod. In simultaneous mode, each device *must* have a unique value.

Double-click any cell in the Amplitude or Number of Periods columns to edit it.

5.2.4 Save Location

Click `Browse...` to choose where scan data files will be saved. Default: current working directory.

5.2.5 Scan Controls

Button	Action
`Start Scan`	Begins the measurement (button disables; Stop Scan enables).
`Stop Scan`	Gracefully stops an in-progress scan and restores nominals.
`Preview Waveform`	Opens a popup window showing the computed sinusoidal waveforms for all devices.
`Save Scan Setup`	Saves all scan parameters to a `.json` file.
`Load Scan Setup`	Restores parameters from a previously saved `.json` file.

5.2.6 What Happens During a Scan

1. FORMA fetches the nominal (current operational) values for all setting devices.
2. For each step $n = 0, 1, \ldots, N_{\text{total}}-1$:
   (a) Computes the excitation value for each corrector using Eq. **??**.
   (b) Applies the settings to hardware via ACNET.
   (c) Waits for the next ACNET event trigger, then reads all device readbacks.
   (d) Stores the timestamp and readback values.
   (e) Checks safety thresholds (if enabled).
3. Progress bar advances with each step.
4. On completion (or abort): restores all correctors to nominal values.
5. Data is auto-saved to CSV with an accompanying `_meta.json` file.

Safety Abort

If the safety system detects an orbit deviation exceeding the abort threshold, the scan halts immediately, nominals are restored, and a detailed abort log is saved. See Section **??** for configuration.

5.3 Tab 3: Safety Configuration

The safety system provides real-time protection during scans.

5.3.1 Safety Status Banner

The banner at the top of this tab shows one of three states:

State	Colour	Meaning
DISABLED	Grey	Safety monitoring is turned off.
NOT CONFIGURED	Yellow	Enabled but no baselines measured yet.
ACTIVE	Green	Fully operational with N devices monitored.

5.3.2 Setup Procedure

Step 1: Enable Safety

Check the box: "*Enable Safety Monitoring (stops scan if thresholds exceeded)*".

Step 2: Measure Baseline

1. Set *Number of Samples* (minimum 10, recommend 50–100).
2. Click `Measure Safety Baseline`.
3. Wait for data collection to complete. The baseline table populates with: Device, Mean, RMS, Std Dev, Min, Max, Samples.

Important

Measure baselines with *no active excitation*. The baseline captures the natural noise floor; any deliberate orbit modulation will inflate the standard deviation and desensitise the safety system.

Step 3: Configure Thresholds

1. **Monitoring Mode**: choose between:
   - **Per-Device Mean** — each BPM checked individually.
   - **Overall Mean** — combined mean across all BPMs.
2. **Averaging Window**: number of recent readings to average before checking (1 = no averaging).
3. **Warning Threshold (σ)**: triggers a log message. Scan continues.
4. **Abort Threshold (σ)**: halts the scan immediately.
5. Click `Apply Threshold Settings`.

5.3.3 Save / Load / Clear

`Save Safety Config`	Exports all baselines and thresholds to a `.json` file.
`Load Safety Config`	Restores a previously saved configuration.
`Clear All Baselines`	Deletes all baseline data (requires confirmation).

5.4 Tab 4: Baseline RMS

A standalone tool for measuring BPM noise floors, independent of the safety system.

1. Set *Number of Samples* (default: 50).
2. Click `Measure Baseline RMS`.
3. Results appear in the table: Device, RMS Fluctuation.
4. Click `Save RMS Data to CSV` to export.

This is useful for characterising machine noise before deciding on scan amplitudes.

5.5 Tab 5: Data Reading and Plots

Provides continuous live monitoring of device readbacks.

5.5.1 Controls

`Start Reading` / `Stop Reading`	Toggles continuous data acquisition.
`Save Live Data`	Saves all accumulated data to CSV.
Device to Plot	Dropdown selecting which device's time series to display.
ACNET Event	Trigger event for data acquisition. Options: `@p,1000` (periodic 1 s), `@i` (injection), `@e,52` (custom event).

5.5.2 Live Plot

The plot shows the most recent 2000 data points for the selected device. The X-axis is timestamp, Y-axis is the readback value. The plot updates in real time during acquisition.

5.6 Tab 6: Statistics

Displays summary statistics for all reading devices during live data acquisition.

Column	Description
Device	Device name.
Min	Minimum observed value.
Max	Maximum observed value.
Mean	Average value.
Std Dev	Standard deviation.

A summary label shows the **Overall RMS of Std Deviations** across all devices. Click `Export to CSV` to save.

5.7 Tab 7: FFT Analysis

An interactive spectral analysis tool for inspecting scan data.

5.7.1 Loading Data

Click `Load Scan Data File` and select a scan CSV. The device listbox populates with all columns from the file. If a corresponding `_meta.json` exists, scan parameters are loaded automatically.

5.7.2 Generating FFT Plots

1. Select one or more devices in the listbox (Ctrl+Click for multiple).
2. Click `Generate FFT Plot`.
3. The plot shows each device's amplitude spectrum.

5.7.3 Plot Features

- **X-axis**: Number of Periods (= number of complete cycles within the scan window).
- **Y-axis**: Normalised Amplitude.
- **Hover**: Moving the mouse over a peak shows an annotation with the exact period count and amplitude.
- **Colour coding**: Each device gets a distinct colour from the palette (cyan, orange, green, yellow, purple, pink).

Verifying Scan Quality

Use the FFT tab to check that:

- Each corrector shows a clear peak at its configured number of periods.
- BPMs show peaks at the corrector frequencies (confirming response).
- The noise floor is well below the signal peaks (good SNR).

5.8 Tab 8: Response Matrix

This is the primary output tab where the ORM is computed and visualised.

5.8.1 Controls

`Calculate Response Matrix`	Runs the full ORM computation (Section ??–??). Requires scan data to be loaded in the FFT tab.
`Save Matrix to CSV`	Exports the matrix to separate CSV files per plane (horizontal/vertical) with corresponding error matrices.
`Open Advanced Analysis`	Launches the PyQt5 analysis viewer (Chapter ??).
Auto-calculate ORM	Checkbox. If enabled, ORM is computed automatically after each scan completes.

5.8.2 Display Modes

The *Display* dropdown switches between three visualisation modes:

Mode	Description
Gain (Real)	The response matrix R_{ij} (Eq. ??). Blue/red coolwarm colourmap; sign indicates kick direction.
Uncertainty (σ)	The error matrix ΔR_{ij} (Eq. ??). Magma colourmap; brighter = larger uncertainty.
Device Noise	Noise floor characteristics.

5.8.3 Heatmap Display

The tab shows two side-by-side heatmaps:

- **Left**: Horizontal BPMs (names containing `BPH`).
- **Right**: Vertical BPMs (names containing `BPV`).

Each has its own colourbar. The X-axis labels are correctors; Y-axis labels are BPMs.

5.8.4 Saved File Format

When saving, FORMA creates up to four files:

- `response_matrix_<timestamp>_horizontal.csv`
- `response_matrix_<timestamp>_vertical.csv`
- `response_matrix_<timestamp>_horizontal_error.csv`
- `response_matrix_<timestamp>_vertical_error.csv`

Each CSV has BPM names as row indices and corrector names as column headers.

5.9 Tab 9: Activity Log

All application messages are displayed here in a scrollable, colour-coded log:

Prefix	Colour	Meaning
[SUCCESS]	Green	Operation completed successfully.
[WARNING]	Orange	Non-critical issue; action may be needed.
[ERROR]	Red	Critical failure; check and resolve.
[INFO]	Grey	Informational message.

Click `Clear Log` to erase the log contents.

Chapter 6

Advanced Analysis Viewer

The analysis viewer is a separate PyQt5 application that provides interactive exploration of the response matrix.

6.1 Launching

There are three ways to open the viewer:

1. **From Forma**: Click `Open Advanced Analysis` in the Response Matrix tab. Data is passed in-memory.
2. **From command line**:

```
python analysis_viewer.py --file response_matrix.csv
```

3. **Programmatically**:

```
from analysis_viewer import launch_from_arrays
proc = launch_from_arrays(matrix, row_labels, col_labels,
                          error_matrix, plane_map)
```

6.2 Window Layout

The viewer has a dark theme matching Forma. The layout is:

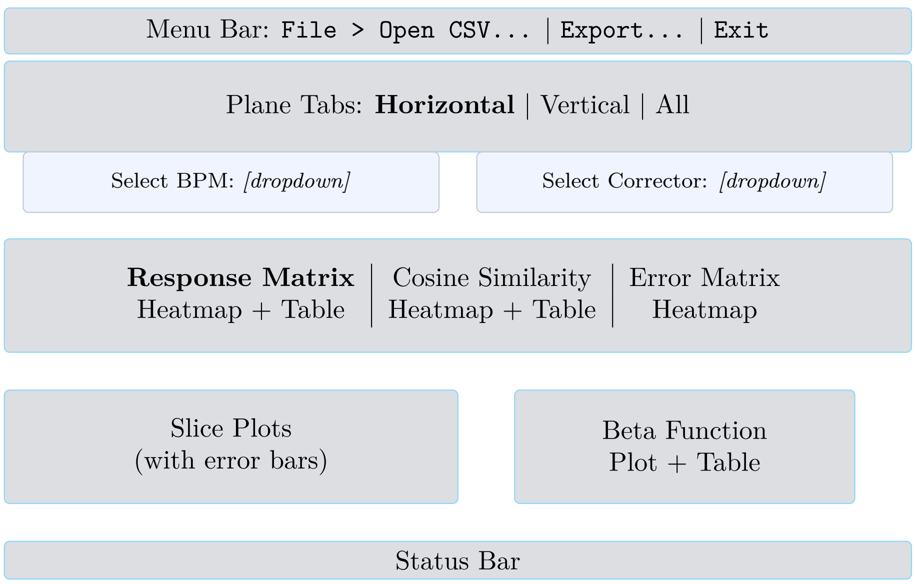

Figure 6.1: Analysis viewer layout.

6.3 Plane Tabs

If a plane map is provided (e.g. from FORMA), separate tabs are created for each plane:

- **Horizontal** — only BPMs containing `BPH` in their name.
- **Vertical** — only BPMs containing `BPV` in their name.
- **All** — full matrix (added if not all rows are covered by the plane map).

Each plane tab is fully independent with its own selectors, plots, and tables.

6.4 Response Matrix Sub-Tab

- **Left panel**: Heatmap coloured with the `viridis` colourmap. Axes show BPM names (rows) and corrector names (columns).
- **Right panel**: Numerical table showing the matrix values to 4 decimal places. Entries that were not computed show "N/A".

6.5 Cosine Similarity Sub-Tab

Shows the corrector-to-corrector similarity matrix (Eq. **??**):

- **Heatmap**: Square matrix with corrector labels on both axes. Diagonal is always 1.0.
- **Table**: Numerical values to 4 decimal places.

Identifying Redundant Correctors

Corrector pairs with $S_{kl} \approx 1$ have nearly identical effects on the orbit. One of them may be redundant for correction purposes.

6.6 Error Matrix Sub-Tab

Displays the uncertainty matrix ΔR_{ij} (Eq. **??**) as a heatmap with the `magma` colourmap (dark = low uncertainty, bright = high). This tab is only enabled if error data was provided.

6.7 Slice Plots

The lower-left panel shows two interactive plots that update when you change the BPM or corrector selection in the dropdowns:

- **Top plot** — BPM Response: for the selected BPM, shows $R_{i,\bullet}$ (response to every corrector). Plotted as cyan circles connected by lines, with error bars (lavender caps) if available.
- **Bottom plot** — Corrector Response: for the selected corrector, shows $R_{\bullet,j}$ (response at every BPM). Plotted as green squares with error bars.

Reading the Error Bars

The error bars represent $\pm\Delta R_{ij}$ from Eq. **??**. A BPM–corrector pair with a large error bar relative to the response value has low measurement confidence and may benefit from:

- Larger corrector excitation amplitude,
- More scan points (better frequency resolution),
- More superperiods (noise averaging).

6.8 Beta Function

The lower-right panel shows the estimated betatron function (Section **??**):

- **Plot**: $\hat{\beta}_i$ vs. BPM index, plotted as orange circles.
- **Table**: Two columns — BPM name and $\hat{\beta}$ value (6 decimal places), or "N/A" if not computable.

6.9 File Menu

`File > Open Response CSV...`	Load a response matrix from a CSV file. First column is the row index (BPM names), header row has corrector names.
`File > Export Current Plane...`	Save the current plane's matrix to a new CSV file.
`File > Exit`	Close the viewer.

Chapter 7

Data Files and Formats

7.1 Scan Data CSV

```
Timestamp, H:CORR01, H:CORR02, BPH:01, BPH:02, ...
2025-01-15 10:30:45.123, 0.50, -0.32, 1.234, 0.567, ...
2025-01-15 10:30:46.135, 0.53, -0.30, 1.245, 0.578, ...
...
```

- First column: ISO-format timestamps.
- Remaining columns: readback values for each device (both correctors and BPMs).
- Correctors use actual readback values, not commanded setpoints.

7.2 Scan Metadata JSON

```
[
  {"device": "H:CORR01", "amplitude": 0.5, "periods": 2},
  {"device": "H:CORR02", "amplitude": 0.3, "periods": 3}
]
```

One entry per corrector. Saved alongside the scan CSV with suffix `_meta.json`.

7.3 Response Matrix CSV

```
Reading Device, H:CORR01, H:CORR02, ...
BPH:01, 0.4532, 0.1234, ...
BPH:02, 0.3812, -0.0567, ...
```

Row index: BPM names. Columns: corrector names. Values: R_{ij}. Missing entries appear as empty cells (NaN).

7.4 Error Matrix CSV

Same format as the response matrix CSV but with ΔR_{ij} values. Saved with suffix `_error.csv`.

7.5 Safety Configuration JSON

Contains: enabled flag, threshold type, per-device and overall warning/abort thresholds, buffer size, and all device baselines (mean, RMS, std, min, max, sample count).

7.6 Scan Setup JSON

Contains: points per superperiod, number of superperiods, scan mode, per-device amplitudes and period counts.

Chapter 8

Tips and Best Practices

8.1 Choosing Scan Parameters

Parameter Selection Guidelines

- **Amplitude**: Large enough for clear signal above BPM noise. Use the Baseline RMS tab (Tab 4) to gauge the noise floor. A rule of thumb: amplitude should produce orbit changes at least 5× the BPM RMS noise.
- **Points per superperiod**: At least 4× the highest number of periods (Nyquist). Recommend 50–200.
- **Number of periods**: In simultaneous mode, assign a *unique* value to each corrector. Avoid integer multiples (e.g. 2 and 4) to prevent harmonic overlap. Good choices: consecutive primes or well-separated integers.
- **Number of superperiods**: More superperiods improve SNR. Use 1–3 for quick checks, 5+ for production measurements.

8.2 Simultaneous vs. Sequential

	Simultaneous	Sequential
Speed	Fast (1×)	Slow (N×)
SNR	Lower (shared bandwidth)	Higher (isolated signal)
Cross-talk	Possible (harmonic overlap)	None
Best for	Routine measurements	Precision measurements

8.3 Interpreting the Response Matrix

- **Large positive/negative R_{ij}**: Strong coupling between corrector j and BPM i. Sign indicates direction.
- **Near-zero R_{ij} with small error bar**: Genuine weak coupling (BPM is near a node of the betatron oscillation).
- **Near-zero R_{ij} with large error bar**: Inconclusive — measurement noise dominates. Consider increasing the scan amplitude or number of superperiods.
- **NaN entries**: Pair was not computed (e.g. missing data or failed FFT). These are displayed as blank cells and transparent pixels in heatmaps.

8.4 Interpreting Error Bars

- Error bars represent $\pm 1\sigma$ uncertainty from Eq. **??**.
- If error bars are much larger than the response value, the measurement is noise-dominated.

- To reduce error bars: increase corrector amplitude (A_c), add more scan points, or add more superperiods.
- The two dominant contributions are:
  1. BPM noise (σ_b): reduced by stronger excitation.
  2. Corrector noise (σ_c): reduced by cleaner power supplies or longer averaging.

8.5 Safety System Recommendations

Safety Best Practices

- Always measure baselines *without* any active excitation.
- Set the warning threshold at 3σ and abort at 5σ as starting values.
- Use the averaging window (buffer size) to smooth transient spikes: 5–20 samples is typical.
- After loading a safety configuration, verify that baselines are still valid for current machine conditions.
- Review abort logs after any safety event to understand what happened.

Appendix A

Quick Reference Card

Action	How
Authenticate	Tab 1 → `Kerberos Login`
Add devices	Tab 1 → Manual entry or `Open Device Selection Tool...`
Preview waveform	Tab 2 → `Preview Waveform`
Run scan	Tab 2 → `Start Scan`
Stop scan	Tab 2 → `Stop Scan`
Setup safety	Tab 3 → Enable → Measure Baseline → Set Thresholds
Measure BPM noise	Tab 4 → `Measure Baseline RMS`
Live monitoring	Tab 5 → `Start Reading`
Inspect FFT	Tab 7 → `Load Scan Data File` → Select devices → `Generate FFT Plot`
Compute ORM	Tab 8 → `Calculate Response Matrix`
View with errors	Tab 8 → `Open Advanced Analysis`
Save ORM	Tab 8 → `Save Matrix to CSV`
Auto-compute ORM	Tab 8 → Check "Auto-calculate ORM after scan"

Appendix B

Key Equations Summary

$$\text{DFT:} \quad X[k] = \textstyle\sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N} \tag{??}$$

$$\text{Amplitude:} \quad A[k] = \tfrac{2}{N}\, |X[k]| \tag{??}$$

$$\text{Transfer function:} \quad R_{ij} = \mathrm{Re}\big(\tilde{A}_{b,i}/\tilde{A}_{c,j}\big) \tag{??}$$

$$\text{Error:} \quad \Delta R_{ij} = \sqrt{\frac{\sigma_b^2}{2A_c^2} + \frac{A_b^2\, \sigma_c^2}{2A_c^4}} \tag{??}$$

$$\text{Beta function:} \quad \hat{\beta}_i = (U_{i,1}\, \sigma_1)^2 \tag{??}$$

$$\text{Cosine similarity:} \quad S_{kl} = \hat{\mathbf{c}}_k \cdot \hat{\mathbf{c}}_l \tag{??}$$

$$\text{Safety deviation:} \quad n_\sigma = |\bar{x} - \mu^{(0)}| \,/\, \sigma^{(0)} \tag{??}$$